The Laplace-transform threshold for $(1 + mse^s)^{-1}$

MathDB problem 350435

12 September 2026

Task 1. Solve the Laplace-transform criterion

Kesidis, Konstantopoulos and Zazanis conjectured the following classification in Conjecture 1 of [1, p. 18]. Their probabilistic construction already established the sufficient range. We prove the missing necessity and give a direct construction for sufficiency.

Theorem 1. *For a real parameter $m \geq 0$, the function*

$$L_m(s) = \frac{1}{1 + mse^s}, \qquad s \geq 0,$$

is the Laplace transform of a probability distribution if and only if $m = 0$ or $m \geq e$. The conclusion holds even if the putative random variable is initially allowed to take negative values.

We use the following elementary form of the nonnegative-coefficient singularity principle, often called Pringsheim's theorem. Its proof is included to make the argument self-contained.

Lemma 1. *Let $f(z) = \sum_{n \geq 0} a_n z^n$, where $a_n \geq 0$, have radius of convergence R with $0 < R < \infty$. Then f has no holomorphic continuation to a neighborhood of R agreeing with the series just to the left of R.*

Proof. Suppose such a continuation exists on $|z - R| < \delta$, where $0 < \delta < R$. Set $b = R - \delta/4$ and $h = \delta/3$. The disc $|z - b| < \delta/2$ is contained in $|z - R| < \delta$. Consequently the Taylor series at b converges at $b + h$. Since $0 < b < R$, termwise differentiation of the original series gives

$$\frac{f^{(k)}(b)}{k!} = \sum_{n \geq k} a_n \binom{n}{k} b^{n-k} \geq 0.$$

All terms in the next double series are nonnegative, so summation can be interchanged:

$$\infty > \sum_{k \geq 0} \frac{f^{(k)}(b)}{k!} h^k = \sum_{n \geq 0} a_n \sum_{k=0}^{n} \binom{n}{k} b^{n-k} h^k = \sum_{n \geq 0} a_n (b + h)^n.$$

But $b + h = R + \delta/12 > R$, contradicting the radius of convergence. □

Proof of the theorem: necessity. Suppose $0 < m < e$ and $L_m(s) = \mathbb{E}e^{-sX}$ for every $s \geq 0$. As $s \to \infty$, $L_m(s) \to 0$. If $\mathbb{P}(X < 0) > 0$, then $\mathbb{P}(X \leq -\varepsilon) > 0$ for some $\varepsilon > 0$, and hence $\mathbb{E}e^{-sX} \geq e^{s\varepsilon}\mathbb{P}(X \leq -\varepsilon) \to \infty$. Thus $X \geq 0$ almost surely.

Differentiating its Laplace transform at $s > 0$ gives

$$(-1)^n L_m^{(n)}(s) = \mathbb{E}[X^n e^{-sX}] \geq 0.$$

Differentiation is justified locally by boundedness of $x^n e^{-cx}$ on $x \geq 0$ for each $c > 0$. The explicit expression for L_m is analytic at zero, so its derivatives extend continuously there. It follows that

$$F(z) := L_m(-z) = \frac{1}{1 - mze^{-z}} = \sum_{n \geq 0} a_n z^n \quad \text{near } 0, \qquad a_n = \frac{(-1)^n L_m^{(n)}(0)}{n!} \geq 0. \tag{1}$$

In particular, $a_0 = 1$ and $a_1 = m > 0$.

For every real $t \geq 0$, the elementary inequality $te^{-t} \leq 1/e$ shows that

$$1 - mte^{-t} \geq 1 - m/e > 0, \qquad 0 < F(t) \leq \frac{1}{1 - m/e}. \tag{2}$$

The displayed quotient therefore provides a holomorphic continuation of the germ in (1) to a neighborhood of every point of the positive real axis. If the Taylor radius in (1) were finite, its positive endpoint would contradict Lemma 1. Thus the radius is infinite. The series then equals the quotient on the positive axis by analytic continuation. Nonnegativity of its coefficients implies

$$F(t) = \sum_{n \geq 0} a_n t^n \geq 1 + mt \qquad (t \geq 0),$$

which contradicts the uniform bound in (2). This excludes the whole interval $0 < m < e$. □

Proof of the theorem: sufficiency. For $m = 0$, take $X = 0$ almost surely. If $m \geq e$, choose $\lambda \in (0, 1]$ so that

$$m = \frac{e^\lambda}{\lambda}.$$

Such a choice exists because e^λ/λ is continuous, decreases from infinity to e on $(0, 1]$, and equals e at 1. Put $p = e^{-\lambda}$. Let Z be exponential with rate λ, let J have distribution

$$\mathbb{P}(J = j) = p(1 - p)^j, \qquad j = 0, 1, \ldots,$$

and let $Y_1, Y_2, \ldots$ have common density

$$q(y) = \frac{\lambda e^{-\lambda y}}{1 - p} \mathbf{1}_{(0,1)}(y).$$

Take all these variables independent and define $T = 1 + Z + \sum_{i=1}^{J} Y_i$. This is a finite nonnegative random variable almost surely. The common transform of the Y_i is

$$\phi(s) = \frac{\lambda(1 - e^{-(\lambda+s)})}{(\lambda + s)(1 - p)}, \qquad s \geq 0.$$

In particular, $(1 - p)\phi(s) \leq 1 - p < 1$, so the geometric series for the transform of the random sum converges. Independence yields

$$\begin{aligned} \mathbb{E}e^{-sT} &= e^{-s} \frac{\lambda}{\lambda + s} \frac{p}{1 - (1 - p)\phi(s)} \\ &= \frac{\lambda e^{-(\lambda+s)}}{s + \lambda e^{-(\lambda+s)}} = \frac{1}{1 + (e^\lambda/\lambda)se^s} = L_m(s). \end{aligned}$$

This includes the endpoint $m = e$, with $\lambda = 1$. □

Scope. The argument establishes the complete parameter classification, not just a finite collection of derivative-sign tests. In fact, the necessity proof shows that for every $0 < m < e$ at least one Taylor coefficient of $L_m(-z)$ is negative. The sufficiency proof constructs an absolutely continuous probability law supported in $[1, \infty)$ for every $m \geq e$.

Task 2. Quantify the first failed derivative sign

For $0 < m < e$, determine or sharply bound

$$n_*(m) = \min\{n \geq 0 : [z^n](1 - mze^{-z})^{-1} < 0\}.$$

In particular, obtain its asymptotic behavior as $m \uparrow e$. The proof above guarantees finiteness but gives no effective bound.

Task 3. Locate the sign changes of the inverse transform

For $0 < m < e$, study the piecewise polynomial inverse Laplace transform of L_m. Determine its first negative interval and obtain rigorous large-time asymptotics, including the transition as $m \uparrow e$.

Task 4. Classify fractional-power variants

For $m, \tau > 0$ and $0 < \alpha \leq 1$, determine precisely when

$$(1 + ms^\alpha e^{\tau s})^{-1}$$

is a probability Laplace transform. Find stochastic constructions throughout the admissible parameter region.

Task 5. Determine the shape of the admissible distributions

For the probability laws T_m, $m \geq e$, determine whether their densities are log-concave and classify the monotonicity of their hazard rates. Obtain sharp bounds that remain valid at the critical value $m = e$.

References

[1] G. Kesidis, T. Konstantopoulos and M. A. Zazanis, *Age of information without service preemption*, arXiv:2104.08050v3 (2021), Conjecture 1, p. 18.